
CASEPATH: AN AGENTIC, PROCESS-FIRST ARCHITECTURE FOR DETERMINING EVIDENCE REQUIREMENTS

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ABSTRACT

Insurance claims are a useful testbed for AI decision systems because the evidence needed to resolve a case depends on where that case lies in a rule-governed process. A single fact can activate a different branch, create new obligations, and change which documents are justified. Yet many LLM-based approaches bypass this structure and predict document checklists directly from case text. Such requests may look reasonable while being weakly grounded: the system can ask for the right-looking document for the wrong reason. We introduce CASEPATH, an agentic, process-first architecture in which the source-defined process—not the language model—controls evidence acquisition. Agents read authoritative sources, place the case on the applicable process path, identify active or unresolved requirements, and request only the evidence needed to satisfy them. As evidence arrives, the case state is updated and the required checklist changes accordingly. We test this dependency using matched cases that differ in one branch-defining fact. Across 27 held-out pairs, CASEPATH makes 15 false-positive document changes, versus at least 27 for every comparison system. This turns document prediction into evidence planning that is selective, traceable, and auditable.

1 INTRODUCTION

A customer reports a burglary. Among the stolen items is a watch that the policy schedules separately at CHF 4,000. Under the applicable household-insurance wording, that single fact activates an obligation that did not exist a sentence earlier: the value of a scheduled valuable must be established, by a purchase receipt or by an expert valuation. Remove the fact and the obligation disappears. Leave it uncertain and the justified move is a question, not a document request. A predictor that maps case text to a list of documents cannot represent this dependency. It can learn that theft claims come with receipts, and it will ask for one whether or not the case justifies it.

CASEPATH is an agentic architecture in which the authoritative source, not the language model, controls evidence acquisition. Its planning object is the chain

source $\rightarrow$ process node $\rightarrow$ obligation $\rightarrow$ required fact $\rightarrow$ evidence capability $\rightarrow$ document route, (1)

and a case’s position on it is a three-valued state: every source condition is true, false or unresolved for the case at hand. Only an active obligation can require a fact, only a required fact can demand evidence, and only a mapped route can name a document. Unresolved conditions produce clarification questions rather than branch-specific requests (Figure 1). When new evidence is admitted, the state is recomputed and the checklist and next action change with it. The model’s roles are semantic (read a passage, decide a condition, judge a document); code owns the consequences, so every request carries the rule, condition and evidentiary purpose that produced it.

Three questions are usually collapsed into one checklist score, and they should not be. *Traceability* asks whether a request has an inspectable chain back to a source. *Branch sensitivity* asks whether changing one case fact changes exactly the documents that fact justifies. *Complete-case utility* asks whether a full plan covers what a case needs without unnecessary demands. None implies the others: a source-linked request can be wrong, a perfect response to one fact can coexist with errors shared by both versions of a case, and a plan can look complete while being unjustified. We evaluate them separately.

A document request depends on the active branch

Source rule: when condition c holds, establish fact f .

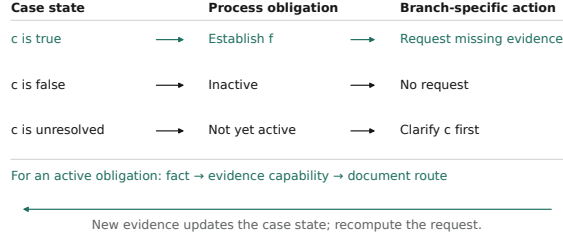


Figure 1: The active branch controls the request. A source condition makes its obligation active, inactive or unresolved. Only an active obligation reaches a required fact, an evidence capability and a document route; an unresolved condition yields a clarification question, never the documents of a branch the case may not be on. New evidence updates the state and the plan is recomputed. Acquisition permission and route sufficiency are specified in Section 2.

Study A tests branch sensitivity with matched interventions. From a frozen snapshot of public Swiss insurance law and household-policy wording we build 36 case pairs whose members differ in one branch-defining statement, with an independently authored reference contract fixing which signed document changes are justified. The benchmark is built to fail input-free shortcuts before any system runs: a constant predictor fitted on the other branch families scores 0.0 on every held-out family (Gururangan et al., 2018; Poliak et al., 2018; Geirhos et al., 2020). On the 27 held-out pairs, CASEPATH makes 15 false-positive signed changes against 27, 41 and 52 for a direct predictor, a graph-as-context predictor and an evidence-first pipeline, with the highest micro- F_1 (0.667), precision (0.615) and exact-pair count (12 of 27). Reassigning its outputs to the wrong branch families collapses F_1 to at most 0.083. The same frozen report rejects the preregistered broad-superiority gate: the F_1 margins over the two strongest comparators are below 0.10 and no family-level test survives Holm correction. We report both: the method’s value is fewer unjustified requests, not a uniformly better checklist.

Study B addresses complete-case utility on the entire released corpus: 150 synthetic tenancy claims in 28 families, 5 learned pipelines sharing one compiled public representation and matched budgets, 2 dependent controls, and native endpoints under a frozen finite-corpus contract. Its development phase is complete and its protected phase is executing at the time of writing (Section 5).

Contributions. (1) An executable evidence semantics: applicability, acquisition permission and route sufficiency as separate three-valued decisions over a closed catalogue, so that no document can be requested without a source chain; evidence ledgers and compiled procedure runtimes already separate semantic judgement from procedural control (Kim, 2026; Yu et al., 2026), and the contribution is the document-planning object and its measured behaviour. (2) A branch-intervention benchmark with built-in falsifiers and a preregistered gate whose failure we report. (3) A matched comparison showing selective, branch-specific and traceable document changes, with the family-level failures that remain. (4) An integrated release: method pack, benchmark, receipts, the 150-claim corpus and contract, and a workbench whose replay agrees with the research planner on all 72 benchmark units.

2 METHOD: SOURCE-GROUNDED PROCESS STATE CONTROLS EVIDENCE

2.1 THE PLANNING OBJECT

A source specification K records process nodes, conditions, obligations, required facts, evidence capabilities and document routes over a closed catalogue $\mathcal{D}$. A capability states what must be shown; a route is a subset of $\mathcal{D}$ that can establish it, and alternative routes remain alternatives. Case state $s = (z, E)$ pairs the assessment z of source conditions with the recorded evidence E . A versioned planner returns the requested documents, a justified next action and the justification chains,

$$(D_t, a_t, J_t) = \Pi_K(s_t), \quad s_{t+1} = U(s_t, e_{t+1}), \quad (2)$$

where U admits a new observation or correction. Two properties hold by construction. *Non-prematurity*: an unresolved condition can emit a question but never a branch-specific document. *Provenance closure*: D_t is a projection of active chains, so no document can enter a plan without an obligation, a fact, a capability and a route behind it.

2.2 AGENTIC ROLES, AND WHERE CODE TAKES OVER

Semantic work is done by model calls in fixed roles: offline, extracting quote-anchored propositions from admitted passages, synthesising the macro process, refining propositions into guarded evidence rules and mapping capabilities onto the catalogue; online, deciding for each guard whether a case satisfies it. Every role returns a structured judgement, never a checklist. Code owns activation, projection, route selection, next-action priority and the justification record (Appendix A, Figure 5): a model may say what a passage means or whether a case meets a condition, but it cannot add a document to the plan.

2.3 STUDY A PLANNER: GUARDED RULES OVER A CLOSED CATALOGUE

The frozen Study A specification was compiled from 197 exact passages of Swiss insurance law and public household-insurance policies and forms (Fedlex, CSS, AXA, Generali, Simpego and the Swiss Insurance Association), with recorded URLs and content hashes. Extraction yields 308 operational propositions. A deterministic check requires every proposition to be either assigned to an evidence rule or explicitly classified as non-evidentiary; the frozen representation has 98 evidence rules, 13 non-evidentiary classifications, 87 case guards and a catalogue of 33 document types. Completeness holds over this extracted inventory only.

Each rule r has a local applicability guard g_r , required capabilities $\mathcal{C}_r$ and, for each capability c , routes $\mathcal{R}_c$. The interpreter assigns $z_g(x) \in \{\text{T}, \text{F}, \text{U}\}$ to guard g for case x . A true or false verdict is accepted only with an exact quotation from that same case; otherwise it reverts to unresolved. Absence of evidence is unresolved, never false. Each guard is decided in its own call, with case packets batched only as an execution optimisation, so every branch decision is auditable on its own. Let $A(x)$ be the unconditional rules and the rules whose guards are true, and let $\mathcal{R}_g^{\text{conf}}$ be the confirmation routes mapped for a guard that is true in the narrative but not yet established by admitted evidence. With held document types H , the plan is

$$D(x, H) = \left[\bigcup_{r \in A(x)} \bigcup_{c \in \mathcal{C}_r} \bigcup_{\rho \in \mathcal{R}_c} \rho \cup \bigcup_{g \text{ true, unconfirmed}} \bigcup_{\rho \in \mathcal{R}_g^{\text{conf}}} \rho \right] \setminus H. \quad (3)$$

An unresolved guard yields a clarification question; a false guard contributes nothing; no generative step runs after the verdicts. Each request retains its rule, fact, must-show statement and source spans. Two boundaries of this version matter for reading the results: Equation 3 unions mapped routes, so V5 neither selects one sufficient alternative nor judges whether a held document is adequate for a specific capability; and the gate is rule-local, so changing macro-process edges while keeping rules and verdicts fixed cannot change D . “Active path” in Study A means the case-conditioned applicability structure, not traversal of every edge.

2.4 STUDY B CONTROLLER: APPLICABILITY, ACQUISITION AND ADEQUACY

The complete-corpus controller compiles three public tenancy templates deterministically into one representation shared by every arm; it performs no model-based source induction. Its assessment separates predicate truth, document presence, individual adequacy and joint adequacy. On the applicability graph,

$$q_s(z) = \phi_s(z) \wedge_3 J_s(\{q_p(z)\}_{p \in \text{pa}(s)}), \quad a_o(z) = q_{\text{scope}(o)}(z) \wedge_3 \phi_o(z), \quad b_o(z) = a_o(z) \wedge_3 \psi_o(z), \quad (4)$$

where J_s is the declared all/any join over parents (roots use the empty all-join, which is true), $\wedge_3$ is false if any input is false, true if all are true and unresolved otherwise, a_o is applicability and ψ_o a separate acquisition permission. Only $b_o = \text{T}$ permits an immediate request. For capability c , a route ρ is satisfied only when its required evidence is present and adequate and the joint assessment is sufficient; write this state $\rho_c(E)$, with missing or inadequate documents $M_{c\rho}$ and unknown presence

or adequacy forming review items V_{cp} . The controller selects

$$\rho_c^* = \arg \min_{\rho \in \mathcal{R}_c}^{\text{lex}} (\mathbf{1}\{\rho_c(E) \neq \mathbf{T}\}, |M_{cp}|, |V_{cp}|, \text{sort}(M_{cp}), \text{id}(\rho)), \quad (5)$$

per capability rather than by global minimisation, and requests $\bigcup_c M_{cp_c^*}$ over eligible obligations. Inactive obligations add no demand; unresolved eligibility produces conditional candidates and questions; an empty route set is an evidence gap, not licence to invent a document. The next action follows a fixed priority: resolve questions, review evidence, request a document, review an active gap, then choose an eligible substantive action. This controller extends the Study A planning object with sufficiency semantics; it is not a repair of V5, and no comparison between the runtimes is claimed (Appendix B).

3 STUDY A: A BENCHMARK ONLY THE BRANCH CAN PASS

Construction. We select 9 branch concepts whose truth changes at least one evidence requirement under the sources: a scheduled valuable at or above the policy threshold, a bicycle among the stolen items, SIM-card theft with misuse, an identifiable third-party causer, a formal expert procedure, a repair above the consent threshold, a written demand with a forfeiture period (*Nachfrist*), recovery of stolen goods, and insurer declination. Each concept is instantiated in four neutral household-theft contexts, giving 36 pairs and 72 case units. The members of a pair differ in one statement: for the watch above, “No stolen item is separately scheduled in the policy with an insured value of CHF 1,000 or more” becomes “A stolen watch is separately scheduled in the policy with an insured value of CHF 4,000”. A reference contract authored from the same sources before any system was run fixes the acceptable signed changes; where the wording admits alternatives (receipt *or* expert valuation) they remain alternatives, and both pre-hidden amendments are preserved with the earlier versions. One context per concept forms the development split (9 pairs, 18 units); the remaining three contexts (27 pairs, 54 units) stayed sealed until the method, benchmark, analysis code and gate were hash-frozen, and were read once. The held-out split therefore tests context robustness for known rule concepts, not unseen-rule or unseen-policy transfer.

Falsifiers built into the benchmark. Static, volume and pairing shortcuts are excluded before evaluation rather than argued away afterwards (Kaushik & Lipton, 2018; Sugawara et al., 2018; McCoy et al., 2019; Bowman & Dahl, 2021; Yuan et al., 2024). (i) *Input-free control*: a constant signed-set predictor fitted on eight branch families and evaluated on the ninth has maximum held-out micro- F_1 0.0; the nine target signatures are distinct (3.17 bits of entropy over 12 signed document atoms). (ii) *Wrong-pairing permutation*: CASEPATH’s outputs are held fixed and reassigned to other branch families under 8 cyclic shifts and 5,000 seeded derangements; a system whose changes are not branch-specific keeps its score. (iii) *One-shot read with a preregistered gate*: eight conditions (Table 4) were fixed before the held-out execution, including a deliberately strong broad-superiority requirement.

Scoring. For a pair (x^-, x^+) , a system’s signed change is

$$\Delta(D) = \{+d : d \in D(x^+) \setminus D(x^-)\} \cup \{-d : d \in D(x^-) \setminus D(x^+)\}. \quad (6)$$

The route-aware scorer compares Δ with every acceptable reference realisation and keeps the best under a frozen lexicographic rule (Appendix A); requesting the union of two alternatives is not credited as correct. True positives, predictions and references are pooled over the 27 pairs into micro precision, recall and F_1 . Spurious changes are unjustified additions or withdrawals; missed changes are required ones not made; an exact pair reproduces the entire signed set. These count changes between two checklists, not total requests in a claim. Uncertainty uses 10,000 whole-family bootstrap draws; pairwise contrasts use the exact $2^9 = 512$ family sign-flip test with Holm adjustment.

Comparators and budget. *Direct* predicts the document set from the case and source material. *Graph as context* additionally receives the induced process and evidence-rule representation but still generates the set in one step. *Evidence-first* derives evidence needs, retrieves twelve passages by keyword matching, and then predicts documents. All arms use the same model setting (openai/gpt-5.6-terra), provider restriction, temperature zero, source snapshot, catalogue and case texts; the held-out receipts record 54, 54, 108 and 87 calls. Inputs and model conditions are matched; compute is not (Appendix A).

Method	Prec.	Recall	F_1	Exact	Pred./Gold	Spurious	Missed
Direct	0.481	0.758	0.588	0.333	52/33	27	8
Graph as context	0.431	0.939	0.590	0.148	72/33	41	2
Evidence-first	0.235	0.485	0.317	0.074	68/33	52	17
CASEPATH	0.615	0.727	0.667	0.444	39/33	15	9

Table 1: **Held-out signed-change evaluation** (27 pairs, 33 reference changes). Exact is the fraction of pairs whose entire signed set is reproduced; Pred./Gold counts predicted changes against reference changes; Spurious and Missed are false-positive and false-negative signed changes. These are errors in changes between two checklists, not requests in a complete claim.

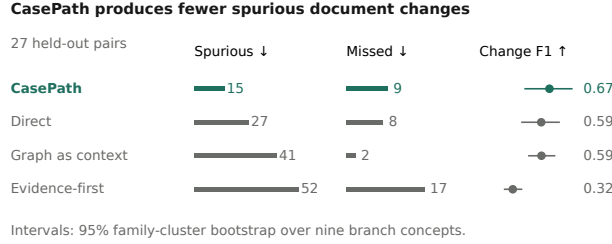


Figure 2: **Fewer unjustified changes.** Spurious and missed signed changes on all 27 held-out pairs, with signed-change F_1 and 95% family-cluster bootstrap intervals over the nine branch concepts. CASEPATH reduces the spurious count; Graph as context recovers more reference changes at the cost of the most spurious ones.

4 STUDY A RESULTS: SELECTIVE AND BRANCH-SPECIFIC, NOT BROADLY SUPERIOR

Fewer unjustified changes. Table 1 and Figure 2 report the single held-out read. CASEPATH predicts 39 signed changes for 33 reference changes: 24 correct, 15 spurious, 9 missed. Direct predicts 52 (25 correct, 27 spurious), Graph as context 72 (31, 41) and Evidence-first 68 (16, 52). CASEPATH thus makes 44.4%, 63.4% and 71.2% fewer spurious changes than the three alternatives while keeping recall 0.727, and it reproduces the whole signed set on 12 of 27 pairs against 9, 4 and 2. Graph as context recovers the most reference changes (31 of 33) at the cost of 41 spurious ones: handing a generator the representation as context makes it ask for more, not for the right things. The contrast with Direct is precise: twelve fewer false-positive changes for one fewer correct change.

The preregistered boundary. Family-cluster intervals are wide: F_1 0.667 [0.435, 0.880] for CASEPATH against [0.404, 0.759] and [0.471, 0.716] for the two strongest comparators, with paired differences $[-0.253, +0.420]$ and $[-0.199, +0.333]$ that include zero; only the contrast with Evidence-first excludes it ($[+0.110, +0.565]$). The broad-superiority gate fails on three of its eight conditions (Table 4): margins +0.078 and +0.076 fall short of 0.10, precision 0.615 falls short of 0.70, and the Holm-adjusted family-swap p -values are 0.797, 0.797 and 0.129. We therefore claim selectivity, not dominance. The failed gate is part of the result, fixed before the read rather than chosen after it.

Branch specificity. Holding CASEPATH’s outputs fixed and pairing them with the wrong branch families collapses signed-change F_1 from 0.667 to a maximum of 0.083 over the 5,008 evaluated reassignments (mean 0.011; add-one randomisation $p = 1.996 \times 10^{-4}$). The changes are specific to the intervened branch. This does not identify the internal representation as the only possible cause; a family-specific lookup would also pass. Two structural facts are separate from accuracy: all 39 of 39 predicted changes are attributed in the retained records to a changed guard, including the incorrect ones, and 27 of 33 reference changes (81.8%) are reachable through a source chain in the frozen representation. Attribution is an invariant of the architecture, not evidence that a mapping was right; reachability says a route existed, not that it was emitted.

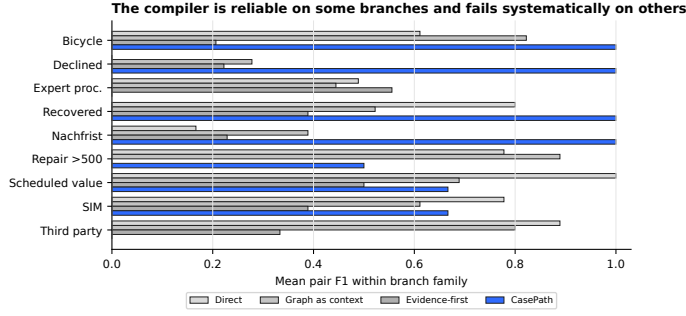


Figure 3: **Where the semantics holds and where it fails.** Mean pair F_1 by branch concept on the held-out contexts. CASEPATH is exact on four concepts and scores zero on two; the comparators are strongest where the plausible claimant document is the right answer.

Where the semantics fails. Figure 3 shows the structure the aggregate hides. CASEPATH is exact on all three held-out contexts of four concepts (bicycle, declination, recovery, *Nachfrist*), reaches 0.667 on scheduled valuables and SIM-card theft, 0.500 on repair consent, and scores zero on third-party causation and the formal expert procedure. The zeros are systematic rather than random: an alternative route is over-requested where one would suffice, a condition is attached at the wrong semantic granularity, or an adjacent procedural guard activates the wrong confirmation document. The comparators are strong where the plausible claimant document is the right answer (third party, repair consent) and weak where the justified change is a procedural artefact that a checklist prior does not suggest (declination, *Nachfrist*); per-concept values are in Appendix A. Finally, a change metric is blind to a document omitted or over-requested in both members of a pair, and these cases are rule units rather than whole claims; Study B therefore scores complete plans on whole cases.

5 STUDY B: THE COMPLETE RELEASED CORPUS

Corpus and arms. Study B evaluates 150 synthetic tenancy claims in 28 families across three domains (defect, mould and heating; lease termination; rent increase). The development split holds 60 claims in 11 families; the protected split holds 90 claims in 17 disjoint families. Five learned arms share one deterministically compiled public representation and the same observable case packet, with 2 calls per cell (draft and review), medium reasoning, 4,096 completion tokens per call, no tools and no resampling, on the same model alias as Study A. CASEPATH predicts an assessment that code compiles through Equations 4–5; the reviewed Direct, Document-first, Process-context and Rule-first arms generate the artefact themselves. Two dependent controls, a compiled equivalent and a local-scope ablation, replay CASEPATH’s assessment through alternative executions, giving 750 learned and 300 dependent cells, 1,050 in all. Because the arms differ in what the model emits, Study B is a pipeline comparison under matched information and budget, not an isolated test of graph storage or prompting order.

Endpoints and contract. The native evaluator scores graph structure, branch and path probes, evidence obligations, chains, document states and exact source locators; a separate endpoint counts the immediate requests a system actually emitted (Appendix C). Three primary endpoints are reported against each learned comparator: emitted-checklist F_1 , critical-evidence recall and the fraction of unnecessary requests. The frozen contract fixes descriptive estimation on the protected split with domains, families and cases weighted equally; conservative contrasts assign -1 whenever either endpoint is unobserved and must jointly exceed the boundaries $+0.03$, -0.03 and -0.03 before a comparator counts as beaten; single-family deletion ranges replace confidence intervals, and no p -values are computed. Failed, truncated or blocked cells stay in the denominator with adverse penalties. All 150 inputs were inspected during product work, so “protected” denotes custody and family disjointness, not untouched-input generalisation.

Execution record. Prediction freezes precede all scoring, and no score exists until both splits are frozen. At the time of writing, 829 of 1,050 cells are recorded: all 420 development cells and 409 of

Arm	Cells of 60			Failure class				
	Valid	Failed	Blocked	Endpoint	Trunc.	Codebook	Presence	Other
CASEPATH	44	16	0	0	2	4	7	3
Direct (reviewed)	19	41	0	21	10	8	0	2
Document-first (reviewed)	17	43	0	13	16	11	0	3
Process-context (reviewed)	27	33	0	26	5	1	0	1
Rule-first (reviewed)	28	32	0	21	8	3	0	0
Compiled-equivalent (dependent)	44	0	16	0	0	0	0	0
Local-scope ablation (dependent)	44	0	16	0	0	0	0	0

Table 2: **Development-split execution record** (420 cells, complete). Failed cells are retained with their charge and enter the final report with adverse penalties. Endpoint: an artefact relation names a node that does not exist; Trunc.: completion limit reached; Codebook: invalid codebook index; Presence: contradictory document-presence state. Dependent controls are blocked when the CASEPATH cell they replay failed.

Missing does not yet mean required.

One recorded prediction - arrears termination with disputed payment timing

SOURCE CONDITION	Arrears	Family home
	Cure notice rule	Spouse notice rule
MODEL STATE	True	Unresolved
OBLIGATION	Applies: true Acquisition: allowed	Applies: unresolved Acquisition: allowed
DOCUMENT STATE	Both documents are missing	
DERIVED PLAN	Request now Cure notice with warning	Keep conditional Clarify family-home status

The request preserves its justification chain

Source rule → applicable obligation → required fact → evidence capability → document

Case cfm_0e11979130681bd0 - 7 immediate requests; 4 conditional - prediction unscored

Figure 4: **Missing does not yet mean required.** One recorded development prediction (the first completed CASEPATH record in sorted cohort order, selected without target access): arrears true, family-home status unresolved, both documents missing. The controller requests the cure notice now and keeps the spouse notice conditional on a clarification; each request preserves its source-obligation-fact-capability-route chain. This shows the executed mechanism, not correctness.

630 protected cells. The completed development phase already exposes a property of the interface (Table 2). Under the same budget, the generative arms fail schema validation in 41, 43, 33 and 32 of 60 cells, against 16 for CASEPATH. Their dominant failure is an artefact relation whose endpoint does not exist (21, 13, 26 and 21 cells), a failure CASEPATH cannot produce because it emits an assessment that code turns into a graph. Truncation at the completion limit follows (10, 16, 5 and 8 against 2). CASEPATH's own failures are contradictory presence states (7) and invalid codebook indices (4). These are execution outcomes, not quality scores, and they penalise every arm alike in the final report.

What the design supports. Figure 4 shows the mechanism on one recorded case. When both freezes are complete, the frozen code produces a single descriptive finite-corpus report, and the release contains the contract, producer, evaluator and every recorded cell so that the report can be regenerated rather than trusted. A positive report would show that explicit obligation execution yields better-justified complete plans than generative pipelines given the same compiled information on this corpus, and nothing about population superiority, mechanism isolation or unseen-input generalisation; it is not pooled with Study A.

6 RELATED WORK

Executable rules and adaptive case state. LegalRuleML represents norms with their source associations (OASIS LegalRuleML Technical Committee, 2021); Catala makes legislative computation executable (Merigoux et al., 2021); DMN formalises decisions and CMMN guard-activated case work (Object Management Group, 2024; 2016); Guard–Stage–Milestone models and DCR reference models of law connect data-driven activation to compliance (Vaculín et al., 2011; López et al., 2020); proactive enforcement derives actions from active obligations (Basin et al., 2024); and insurance software conditions document lists on service definitions (ServiceNow, 2026). These are prior art; CASEPATH adds their explicit coupling to language-model assessments and a measurement of the resulting evidence-planning errors.

Procedure extraction and process-guided agents. PAGED extracts procedural graphs from documents (Du et al., 2024); BREX benchmarks business-rule flows with executable grounding (Yang et al., 2026); Guideline2Graph compiles clinical guidelines into provenance-preserving decision graphs (Kilic et al., 2026); Graus (2026) evaluate structured representations for legal decision models; CANONIC frames governance as compilation (Hadley, 2026). FlowBench and WorFBench evaluate workflow-guided planning and workflow generation (Xiao et al., 2024; Qiao et al., 2025); Procedural Graphs localise an agent to an active node (Lu et al., 2026); ReAct interleaves reasoning and action (Yao et al., 2023). None scores whether a changed case fact changes exactly the justified evidence.

Obligation-controlled acquisition. The closest precedents are Kim (2026), whose evidence ledger activates branch-specific knowledge obligations while a model judges semantic warrant and code manages closure, and Yu et al. (2026), who compile SOP constraints into verification routines with evidence-bearing returns and an active execution frame. CASEPATH’s residual is the document-planning object: applicability separated from acquisition permission, presence from capability-specific and joint adequacy, alternative routes kept as alternatives, and a benchmark that tests the consequence downstream. Neither system is reproduced as an arm, and no superiority over them is claimed.

Grounding, evidence and information seeking. GraphCompliance aligns policy and context graphs for regulatory assessment (Chung et al., 2025); CLER scores litigation evidence lists and the facts they prove (Bi et al., 2026); InfoGatherer selects retrieval and questions by evidential uncertainty (Taranukhin et al., 2026); verified quotation and chain-of-verification make support inspectable (Menick et al., 2022; Dhuliawala et al., 2024). Tool-use benchmarks score whether the right function was called (Li et al., 2023; Patil et al., 2025), the analogue of asking for a plausible document; CASEPATH asks whether the call was justified by the case.

Benchmark validity. Partial-input baselines, annotation artefacts and shortcut learning motivate the input-free control and the wrong-pairing permutation (Kaushik & Lipton, 2018; Poliak et al., 2018; Gururangan et al., 2018; McCoy et al., 2019; Sugawara et al., 2018; Geirhos et al., 2020; Bowman & Dahl, 2021; Yuan et al., 2024). Study A’s references are authored from sources rather than adjudicated by a model, avoiding the preference leakage that arises when judged and judging models coincide (Li et al., 2026); work on evaluation validity (Thomas et al., 2026) and critiques of legal benchmarks that reward outcome prediction over practitioners’ work (Medvedeva & McBride, 2023; Lou & Shin, 2026) motivate signed document changes and emitted requests as the measured objects.

7 RELEASE: ONE PLANNER FOR RESEARCH AND PRODUCT

A research method and a claims interface can silently diverge. The Study A release packages the frozen representation, propositions, snapshot, method manifest, benchmark identity and model configuration into a hash-bound method pack; the client cannot inject rules, mappings or targets, and hash drift makes the endpoint fail closed. The product calls the same guard interpreter and planner as the experiment and exposes guard states, chains, questions, gaps, requests, exact source spans and the next action. Replaying all 72 benchmark units with the recorded guard decisions reproduces the

research document sets, guard states and next actions on every unit and all 36 signed pair changes, with every rendered quotation an exact substring of its passage: implementation identity conditional on recorded assessments, not fresh-inference agreement or deployment evidence. The workbench organises six roles (Facts, Orchestration, Source Integrity, Process, Evidence, and Audit and Readiness) around one case record, and a newly admitted observation triggers recomputation through Equation 2. The release also contains benchmark versions, predictions, receipts, analysis code, the parity verifier, and Study B’s corpus, producer, evaluator, contract and every recorded cell.

8 LIMITATIONS

Study A federates statute wording with several insurers’ policies; a pair tests one rule consequence, and deployment requires selecting the governing policy first. The reference contract is authored from sources and mechanically checkable for spans and routes, but which documents operationalise a fact is a modelling judgement, not expert consensus. Held-out contexts cover known concepts only, and no case-only ablation isolates the necessity of source text at inference time. Spurious signed changes proxy unnecessary acquisition rather than measured claimant burden. The V5 planner unions routes and does not judge adequacy; the Study B controller does, but its comparison is pipeline-level, its inputs were previously inspected, and its finite-corpus report supports no population inference. Neither study measures autonomous acquisition, expert agreement or customer outcomes.

9 CONCLUSION

CASEPATH makes one distinction executable: uncertain applicability triggers state resolution, and only active, source-backed obligations generate evidence requirements. Under matched interventions this yields document changes that are selective (15 spurious against 27, 41 and 52), branch-specific (wrong pairing collapses F_1 from 0.667 to at most 0.083) and traceable, while a preregistered gate rightly denies broad superiority and the family analysis names the semantics that still fail. The complete-corpus study, the method pack and the workbench turn document prediction into evidence planning that can be inspected, replayed and falsified.

REPRODUCIBILITY STATEMENT

The release contains, for Study A, the frozen snapshot, propositions, rules, document map, interpreter, planner, benchmark versions, all predictions, provider receipts, analysis code, the preregistered gate and the product replay verifier, and for Study B its corpus, compiled representation, producer, evaluator, analysis contract and every recorded cell, including failures. Every number in this paper is generated from those files by a script that records the file hash and JSON pointer behind each value, and a single command in the release recomputes the Study A scores from the recorded predictions of all four arms and diffs them against the preserved report without network or provider access. Reproducing scores from recorded outputs is distinct from regenerating model outputs, which requires provider access and is not deterministic.

ETHICS STATEMENT

CASEPATH is a research architecture for evidence planning, not legal advice or an automated coverage decision. The studies use public-source snapshots and synthetic cases with no personal data. Real deployment requires policy and jurisdiction scoping, professional review, privacy controls and a mechanism for correction and appeal.

AI USE STATEMENT

Generative AI tools were used in hypothesis formulation, methodological critique, literature discovery, implementation and testing, synthetic-case construction, diagnostic interpretation, and drafting and editing of text and figures. The reproducibility materials distinguish actual executions, engineering fixtures, archived reports and author judgements; failed results and version differences are

retained. The authors are responsible for the final scientific claims, citations, measurements and artefacts.

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A STUDY A: BENCHMARK AND ANALYSIS DETAILS

Branch concepts. The 9 concepts are: a scheduled valuable at or above the source-defined threshold; a bicycle or e-bike among the stolen items; SIM-card theft with a misuse claim; an identifiable third-party causer; selection of formal expert proceedings; contemplated repairs above the consent threshold; a written demand with a forfeiture warning and period; recovery of, or news about, stolen items; and insurer declination. Each has four neutral contexts; a pair changes only the branch statement.

Signed-change scoring. For pair i the scorer evaluates each acceptable signed reference realisation $G \in \mathcal{G}_i$ against the prediction $P_i = \Delta(D_i)$ and selects the maximum of the lexicographic tuple $(F_1(P_i, G), |P_i \cap G|, \text{Prec}(P_i, G), \text{Rec}(P_i, G), -|G|, \text{sort}(G))$, whose last component is the frozen deterministic tie-break. Precision and recall use zero for empty denominators; F_1 is one when prediction and reference are both empty. Micro scores pool the selected realisation counts across the 27 pairs; family means average pair F_1 within each of the nine concepts. The family bootstrap resamples nine whole families with replacement on each of 10,000 draws (seed 20260917),

CasePath compiles source-grounded process state into auditable evidence requests



active process state → obligation → required fact → evidence capability → document route

Unresolved branches become questions; they do not leak branch-specific documents into the checklist.

Figure 5: **The frozen Study A pipeline.** Admitted authority spans are extracted into operational propositions, organised by a macro process, refined into guarded evidence rules and mapped onto a closed document catalogue. At case time only the guards are decided by the model; the plan is a deterministic projection of the active chains.

Benchmark construction blocks static, volume, and pairing shortcuts before evaluation



- ✓ No-input cross-fit: 0.0 F1
- ✓ One-line edit per pair
- ✓ 9 branch families
- ✓ Closed 33-document vocabulary
- ✓ Alternative routes preserved
- ✓ One-shot hidden read

Figure 6: Benchmark construction and admission controls. Held-out contexts remained sealed until the method, benchmark, statistics and runners were hash-frozen.

retaining all three held-out contexts of each selected family. The exact family-swap test enumerates 512 sign assignments of the nine family mean differences; the three one-sided p -values (0.436, 0.398, 0.043) receive Holm adjustment (0.797, 0.797, 0.129). The wrong-pairing diagnostic uses 8 cyclic shifts and 5,000 sampled derangements with add-one score $(1 + N_{\geq})/(1 + 5,008)$; the reported maximum is over this evaluated collection, not all derangements.

Per-concept results. Table 3 lists the held-out mean pair F_1 for every branch concept and arm.

One branch resolution changes the checklist because the active source rule changes

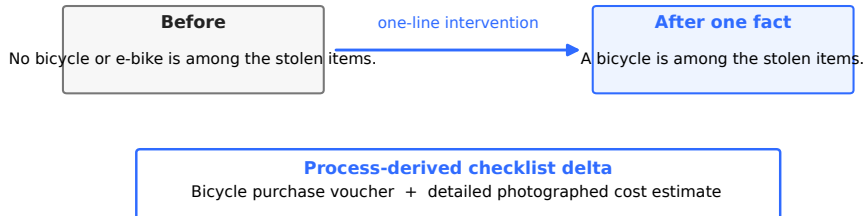


Figure 7: A benchmark intervention changes one process fact. The expected signed checklist change is derived from the source contract, not from any system’s output.

Branch concept	Direct	Graph ctx.	Evid.-first	CASEPATH
Bicycle among stolen items	0.611	0.822	0.206	1.000
Insurer declination	0.000	0.278	0.222	1.000
Formal expert procedure	0.489	0.444	0.556	0.000
Recovery of stolen goods	0.800	0.522	0.389	1.000
Written demand with period (Nachfrist)	0.167	0.389	0.229	1.000
Repair above consent threshold	0.778	0.889	0.000	0.500
Scheduled valuable	1.000	0.689	0.500	0.667
SIM-card theft	0.778	0.611	0.389	0.667
Identifiable third-party causer	0.889	0.800	0.333	0.000

Table 3: Held-out mean pair F_1 by branch concept.

Preregistered gate. Table 4 lists the eight conditions fixed before the one-shot read and their outcomes.

Preregistered held-out gate	Result
F_1 margin ≥ 0.10 against every comparator	fail
Holm-adjusted family-swap $p \leq 0.05$ against every comparator	fail
Signed-change precision ≥ 0.70	fail
Predicted/gold change ratio in $[0.75, 1.50]$	pass
Predicted changes attributed to a changed guard ≥ 0.90	pass
Reference changes reachable through a source chain ≥ 0.80	pass
Correct pairing above the wrong-pairing 97.5th percentile	pass
No requested document without a source chain	pass

Table 4: Preregistered held-out gate, fixed before the one-shot read. The provenance, volume and pairing conditions pass; the broad-superiority conditions do not.

Frozen record and development separation. The method freeze `casepath.theft-method-freeze/5.0.0` binds source bytes, benchmark v3 and both amendments, the proposition-complete refinement, document mapping, interpreter V4, planner V3, model, provider and temperature settings, analysis code, the development split, the matched comparators’ raw predictions and the focused method tests. Development used 9 pairs (18 units), one context per concept, and its pre-declared admission checks authorised exactly one held-out read of the remaining 27 pairs (54 units) under a separately hash-bound execution freeze. After the held-out result, the method was marked locked and no scientific file was adapted from held-out outcomes.

Calls and recorded costs. Table 5 lists the held-out provider receipts per arm.

Arm	Calls	Prompt tokens	Completion tokens	Max completion	Cost (USD)
Direct	54	2,900,625	45,678	14,000	4.83
Graph as context	54	5,345,421	42,663	14,000	8.41
Evidence-first	108	259,890	66,450	14,000/18,000	1.06
CASEPATH	87	1,005,211	243,891	10,000	5.44

Table 5: Held-out provider receipts, one logged attempt per call, model `openai/gpt-5.6-terra`, temperature 0. Evidence-first allows 18,000 completion tokens for extraction and 14,000 for planning; CASEPATH allows 10,000 per guard batch; Direct and Graph as context allow 14,000 per call. Costs are provider-reported charges, exclusive of source preparation; the arms are not compute-matched.

Product-parity contract. The parity verifier loads the installed hash-bound method pack and the real product service, replays all 72 units with the recorded guard decisions, compares guard verdicts, document sets and next actions, compares all 36 pair deltas through the product diff surface, and

Binding	Study A: theft-method-freeze/5.0.0	Study B: native-150 compact controller
Knowledge	Source-extracted propositions, macro graph, refined local evidence rules and mapped routes	Common deterministic compilation of all three public tenancy templates
Online gate	Rule-local true/false/unresolved guard; macro edges do not gate requests	Inherited applicability scopes, obligation condition and separate acquisition permission
Evidence state	Held document types; true-guard confirmation; union of mapped routes	Presence, capability/route adequacy, multi-document joint adequacy; per-capability route selection
State update	Recompute requests/next action from a changed packet; paired edit, not acquired evidence outcome	Recompute from a new assessment; benchmark measures one planning realization
Population	9 development + 27 held-out pairs; nine shared concepts, four contexts; 72 units total	60 development/11 families + 90 protected/17 families; 150 claims, three tenancy domains
Learned inference	Guard-major batched V5; 1-call Direct, 1-call representation baseline, 2-call Evidence-first	Five 2-call singleton pipelines; two additional controls reuse CasePath state
Endpoint	Route-aware signed additions/withdrawals, 27-pair micro scores	Complete native artifact and separate actual immediate-checklist metrics
Analysis	Historical family bootstrap, exact family swaps and wrong-family diagnostic; broad gate failed	Equal domain/family/case finite contrasts; family-deletion sensitivity, no significance claim
Parity	Recorded-answer replay over 72 units and 36 deltas	Separate implementation; no inherited V5 or six-role/live-app parity claim

Table 6: A common source-to-state-to-evidence interface does not make the two implementations, populations or measurements interchangeable. Study B extends the evidence question rather than retrospectively re-evaluating V5.

checks that every rendered quotation is an exact substring of its source passage. A successful HTTP response or a plausible demo is not treated as evidence of implementation identity.

B VERSIONED CORRESPONDENCE BETWEEN THE TWO STUDIES

The shared interface is source specification, case assessment, evidence planning, state replacement and a justified next action. It does not assert identical knowledge, algorithms, estimands or validation across the two implementations. Study A is bound to `casepath.theft-method-freeze/5.0.0`, interpreter V4 and evidence overlay V3; Study B is bound to the compact native producer and its `obligation_control_v1` and `evidence_demand_v1` controller. The native public template union is not a transfer of the theft source pack, and the theft pairs are not transformations of the native claim roster. V5 judges local guards and projects every mapped route on active rules plus mapped confirmation evidence for true unconfirmed guards, minus held document types; it implements neither capability-specific nor joint adequacy. The native controller distinguishes applicability from acquisition, unknown from missing evidence, and per-capability route selection from route union. These are substantive version differences, and no benchmark comparison between the runtimes is claimed. Study A’s preparation extracted propositions, synthesised and refined rules and mapped capabilities with model calls and retained source-only corrections; its reported held-out costs exclude that preparation. Study B’s compilation is deterministic and shared by all arms.

C STUDY B: NATIVE MEASUREMENTS AND ANALYSIS CONTRACT

Native endpoints. Native critical-evidence recall is criticality-weighted coverage of active evidence obligations through any complete permitted document alternative. Evidence-obligation completeness additionally requires valid decision–fact–capability–document relations. Valid-chain precision counts requested items with such a chain over all requested items; its empty-set value is

one and must be read beside coverage. Wrong-branch and premature requests, document states, node and edge scores, branch and path probes and source locators retain their native contracts; locator agreement is not semantic entailment, and probe accuracy is not a direct measure of inferred case-predicate accuracy. The emitted endpoint counts unique immediate request identities without activating candidate conditions from the reference scenario, and is available only when the reference has a unique allowed document set for every active obligation, complete consistent document states and the public label mapping. For predicted and reference sets P, G : $F_1 = 2|P \cap G|/(|P| + |G|)$ and unnecessary fraction $U = |P \setminus G|/|P|$; both-empty F_1 , empty-prediction precision and empty-reference recall are one, and empty-prediction burden is zero. Failed cells use adverse frozen unit penalties instead: zero for higher-is-better rates, one for lower-is-better rates. Unavailable reference labels never become empty sets or silently shrink a denominator.

Estimation. For a split or the complete corpus, each scheduled case has weight $\omega_i = 1/(3 m_{d(i)} n_{f(i)})$, where m_d is its domain’s family count and n_f its family’s scheduled case count. Conventional differences use adverse-imputed unit scores; conservative benefit-oriented differences contribute -1 whenever either producer endpoint is unobserved. Every single-family deletion is reported with remaining weights renormalised within the affected domain; these ranges are composition sensitivity, not confidence intervals. A comparator is counted as beaten only when all three conservative contrasts strictly exceed their boundaries on the complete protected population under the original references and integrity checks; otherwise the report labels the outcome as not met, a partial practical signal, or not evaluable.

Execution and failure classes. Each cell records its arm, case, state and either the validated artefact or the retained provider output with the reason validation failed. Unknown relation endpoint: a candidate artefact relation references a node identity that does not exist in the artefact. Truncated or non-terminal: the completion reached the 4,096-token limit or ended without a terminal outcome. Invalid codebook index: a reference into the shared public codebook that does not resolve. Contradictory presence state: a document marked both present and absent, or present with an inconsistent state. Blocked dependency: a dependent control whose CASEPATH source cell failed. Every failure retains its charge; a repaired version can never replace a failed cell in the frozen comparison.